



Report 9

Echo Bridge Support Network Design Hackathon
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**NETWORK SECURITY –
BATCH 2**



1. Introduction:

The Echobridge Hackathon project was designed to create a reliable and cost-effective enterprise network for a business environment. The main requirement was to provide low-latency voice communication and central server/database access for 24 agents and 1 manager.

The project also had a strict budget limit of 1.5 million PKR, so the network had to be designed carefully without unnecessary hardware or infrastructure.

The network was implemented and tested in Cisco Packet Tracer. VLANs were used to separate different types of traffic, while router-on-a-stick was used for communication between VLANs. Static routing was used to provide connectivity between the different network sections.

The project also used daisy-chaining between IP phones and PCs to reduce the number of physical cable runs and help keep the overall design within budget.

2. Problem Statement

The business required a network that could support:

- 24 agents
- 1 manager
- Reliable voice calls
- Central database/server access
- Separation of voice and data traffic
- Inter-VLAN communication
- Communication between the user network and server network
- A simple and reliable network architecture
- A strict budget of 1.5 million PKR

The main challenge was to provide reliable voice and data services while keeping the infrastructure cost below the given budget.

3. Project Objectives:

The main objectives of this project were:

1. Design an enterprise network for the Echobridge environment.
2. Support 24 agents and 1 manager.
3. Provide voice communication using IP phones.
4. Provide data connectivity for agent PCs.
5. Separate voice and data traffic using VLANs.
6. Provide access to the central server.
7. Configure inter-VLAN routing.
8. Configure routing between different network sections.
9. Reduce infrastructure cost using daisy-chaining.
10. Test and verify the complete network in Cisco Packet Tracer.

4. Business Requirements:

The client requirement was mainly focused on two things:

Voice Communication

The agents required reliable and low-latency voice communication. Voice traffic should remain separated from normal PC data traffic so that heavy data traffic does not negatively affect voice communication.

Central Server Access

The agents also needed access to a central server/database located in the server network.

The design therefore needed to provide both services while maintaining network separation and proper routing.

5. Budget Constraint

The project had a strict budget of:

1.5 Million PKR

The network design could not exceed this budget.

A major cost-saving technique used in the project was **daisy-chaining the PC through the IP phone**.

Instead of providing a separate physical cable for every PC and phone connection, the PC can connect through the IP phone. This reduces the number of physical cable runs and helps reduce infrastructure requirements.

According to the project brief, using separate physical cable runs could require approximately 50 cable runs, while the daisy-chain approach reduces this to approximately 25 cable runs.

This helped the proposed design remain under the budget.

6. Network Design Overview:

The network was divided into different logical sections using VLANs.

The main VLANs defined in the project were:

VLAN	Purpose	Network	Gateway
Vlan 10	Data	192.168.10.0/24	192.168.10.1
Vlan 20	Voice	192.168.20.0/24	192.168.20.1
Vlan 30	Servers	10.1.1.0/24	10.1.1.1

11. Physical Topology

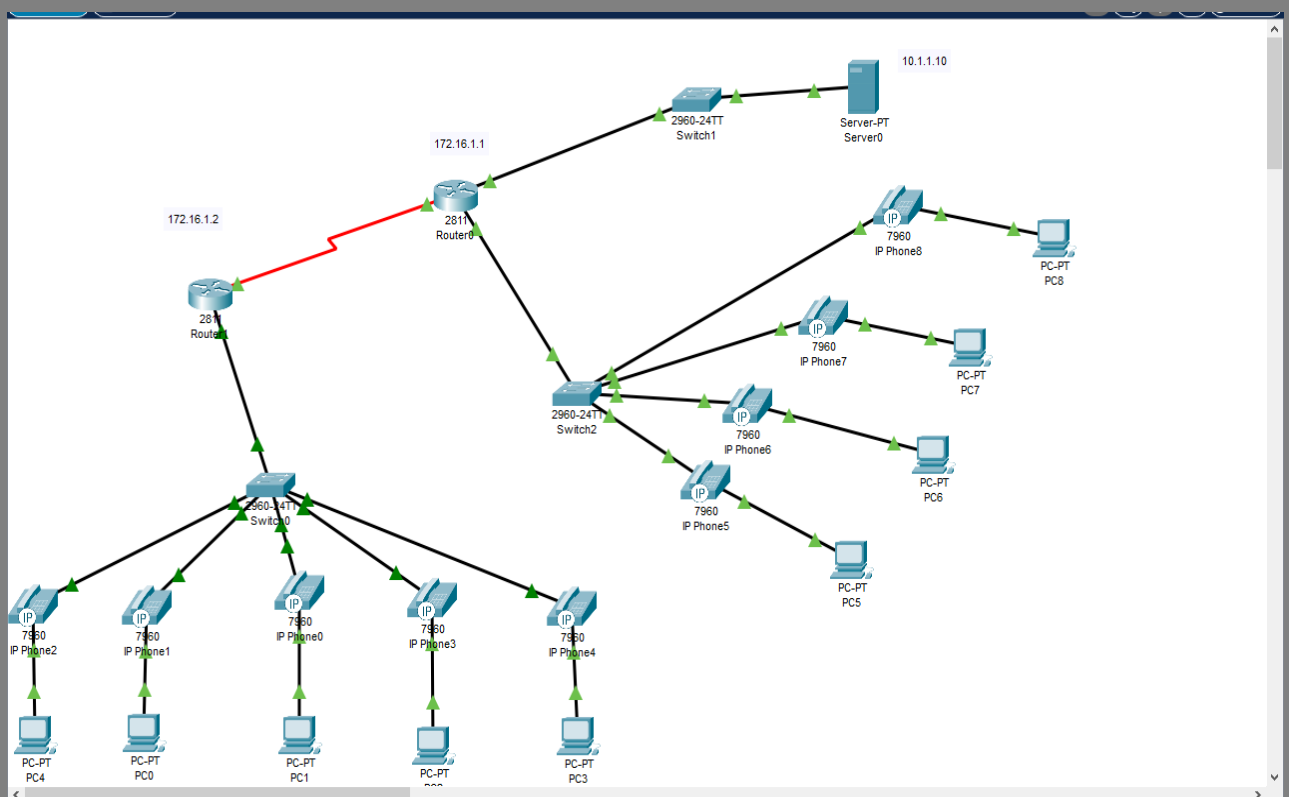
The physical topology consists of routers, switches, IP phones, PCs and the central server.

The PCs and IP phones were connected using the daisy-chain approach.

The basic connection is:

Switch → IP Phone → PC

This approach reduces the requirement for separate switch ports and physical cable runs.



12. Daisy-Chaining

Daisy-chaining was one of the important design decisions of this project.

Instead of connecting every PC directly to a separate switch port, the PC can be connected through the IP phone.

The connection becomes:

Switch → IP Phone → PC

This reduces the number of required physical cable runs.

According to the project brief:

- Separate connections could require around **50 cable runs**.
- Daisy-chaining reduces this to around **25 cable runs**.
- This helps the network remain within the **1.5 million PKR budget**.

13. Access Ports:

Access ports are used to connect end devices to the switch.

Agent PCs are placed in the data VLAN.

IP phones are assigned to the voice VLAN.

This allows the switch to logically separate the traffic even though the PC and IP phone may use the same physical switchport.

14. Trunk Port:

A trunk link is used between the switch and router.

The trunk allows multiple VLANs to travel through the same physical connection.

In this project, the trunk carries:

- VLAN 10 – Data
- VLAN 20 – Voice
- VLAN 30 – Server traffic where applicable to the topology

The trunk is necessary because the router needs to receive traffic belonging to different VLANs.

15. Router-on-a-Stick

Router-on-a-stick was used to provide inter-VLAN routing.

Instead of using a separate physical router interface for every VLAN, one physical router interface is divided into multiple logical sub-interfaces.

Each sub-interface represents a different VLAN.

For example:

Router Interface → Sub-interface → VLAN

The router can therefore route traffic between VLAN 10 and VLAN 20 and between the user networks and the server network.

16. Network Verification:

After configuring the network, connectivity must be tested.

The project instructions specifically require verifying communication from the agent PC to the central server and checking the voice VLAN status on switchports.

The following tests were performed:

Test 1 – PC to Gateway

The agent PC was tested against:

192.168.10.1

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.10.1

Pinging 192.168.10.1 with 32 bytes of data:

Reply from 192.168.10.1: bytes=32 time=1ms TTL=254
Reply from 192.168.10.1: bytes=32 time=16ms TTL=254
Reply from 192.168.10.1: bytes=32 time=1ms TTL=254
Reply from 192.168.10.1: bytes=32 time=26ms TTL=254

Ping statistics for 192.168.10.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 26ms, Average = 11ms

C:\>|
```

23. PC to Server Test

The agent PC was tested for connectivity with the central server.

Traffic path:

Agent PC → VLAN 10 → Router → Server Network → Server

```
C:\>ping 10.1.1.10

Pinging 10.1.1.10 with 32 bytes of data:

Reply from 10.1.1.10: bytes=32 time=18ms TTL=126
Reply from 10.1.1.10: bytes=32 time=1ms TTL=126
Reply from 10.1.1.10: bytes=32 time=1ms TTL=126
Reply from 10.1.1.10: bytes=32 time=13ms TTL=126

Ping statistics for 10.1.1.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 18ms, Average = 8ms

C:\>
```

24. Voice VLAN Verification:

The switchport configuration was checked to verify that the IP phone was associated with the voice VLAN.

The purpose of this verification was to ensure that:

- The phone receives voice VLAN connectivity.
- PC traffic remains in the data VLAN.
- The switchport is correctly configured.

25. Routing Verification:

Routing configuration was checked to make sure that the different network sections could communicate.

The routing table should contain the required networks and paths.

```
R0#sh ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       10.1.1.0/24 is directly connected, FastEthernet0/0
L       10.1.1.1/32 is directly connected, FastEthernet0/0
172.16.0.0/16 is variably subnetted, 2 subnets, 2 masks
C       172.16.1.0/30 is directly connected, Serial0/3/0
L       172.16.1.1/32 is directly connected, Serial0/3/0
192.168.10.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.10.0/24 is directly connected, FastEthernet0/1.10
L       192.168.10.1/32 is directly connected, FastEthernet0/1.10
192.168.20.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.20.0/24 is directly connected, FastEthernet0/1.20
L       192.168.20.1/32 is directly connected, FastEthernet0/1.20
```

28. Scalability:

The /24 networks provide many available IP addresses compared with the current requirement of 24 agents and 1 manager.

This means additional devices can be added in the future without immediately redesigning the addressing scheme.

The architecture can therefore support future expansion of the business.

29. Reliability:

The network was designed to provide reliable connectivity for both voice and data services.

Reliability is improved by:

- Separating voice and data traffic.
- Using structured IP addressing.
- Using proper VLAN configuration.
- Using routing between network segments.

30. Cost Optimization

Cost optimization was an important part of the project.

The client had a strict:

1.5 million PKR budget

The design therefore avoided unnecessary infrastructure.

The main cost-saving technique was daisy-chaining.

Instead of requiring separate physical cable runs for every PC and IP phone, the PC connects through the IP phone.

This reduces:

- Cable requirements
- Required switch ports
- Installation complexity
- Infrastructure cost

The project brief identifies this approach as the method used to keep the design under budget.

31. Configuration Screenshots:

VLAN Configuration:

```
Switch#show vlan brief
```

VLAN	Name	Status	Ports
1	default	active	Fa0/2, Fa0/7, Fa0/8, Fa0/9 Fa0/10, Fa0/11, Fa0/12, Fa0/13 Fa0/14, Fa0/15, Fa0/16, Fa0/17 Fa0/18, Fa0/19, Fa0/20, Fa0/21 Fa0/22, Fa0/23, Fa0/24, Gig0/1 Gig0/2
10	DATA	active	Fa0/3, Fa0/4, Fa0/5, Fa0/6
20	VOICE	active	Fa0/3, Fa0/4, Fa0/5, Fa0/6
1002	fddi-default	active	
1003	token-ring-default	active	
1004	fddinet-default	active	
1005	trnet-default	active	

```
Switch#
```

Trunk Configuration:

```
Switch#show int trunk
Port      Mode      Encapsulation  Status      Native vlan
Fa0/1     on        802.1q         trunking    1

Port      Vlans allowed on trunk
Fa0/1     10,20

Port      Vlans allowed and active in management domain
Fa0/1     10,20

Port      Vlans in spanning tree forwarding state and not pruned
Fa0/1     10,20

Switch#
```

Router Sub-Interface Configuration:

```
R0#show ip int brief
Interface      IP-Address      OK? Method Status      Protocol
FastEthernet0/0 10.1.1.1        YES manual up          up
FastEthernet0/0.10 unassigned      YES unset  administratively down down
FastEthernet0/0.20 unassigned      YES unset  administratively down down
FastEthernet0/1 unassigned      YES manual up          up
FastEthernet0/1.10 192.168.10.1    YES manual up          up
FastEthernet0/1.20 192.168.20.1    YES manual up          up
Serial0/3/0      172.16.1.1      YES manual up          up
Serial0/3/1      unassigned      YES unset  administratively down down
Vlan1           unassigned      YES unset  administratively down down
R0#
```

39. Design Explanation for Business Owner

When presenting this project to a non-technical business owner, the network should not be explained mainly through commands and protocols.

The focus should be on business benefits.

"Our network separates phone communication from normal computer traffic so that important calls are not affected by heavy computer usage. Employees can also access the central server, and we reduced the infrastructure cost by connecting the PCs through the IP phones instead of requiring a separate cable for every device."

The project brief specifically instructs teams to explain why the design solves the business problem instead of focusing on technical commands.

44. Conclusion

The Echobridge Hackathon project successfully demonstrates the design of a cost-effective enterprise network for 24 agents and 1 manager.

The network uses VLAN 10 for data, VLAN 20 for voice, and VLAN 30 for servers. Router-on-a-stick provides inter-VLAN routing, while static routing provides communication between the required network sections.

The most important design decision was the use of daisy-chaining, where the PC is connected through the IP phone. This reduces the number of physical cable runs and helps keep the network within the strict 1.5 million PKR budget.

The network also separates voice and data traffic using VLANs, allowing the business to maintain organized communication between IP phones, PCs and the central server.

The final design was verified through connectivity testing, routing verification and voice VLAN verification.

Overall, the project demonstrates how networking technologies can be used not only to provide connectivity but also to solve real business problems such as cost reduction, traffic separation, centralized access and reliable communication.